

Exam on Dynamical Systems, June 10, 2020

1. (1p=0.2+0.4+0.4) Find the linear homogeneous difference equation with constant coefficients of minimal order that has as solutions the following sequences.

(a) $-3, -6, -12, -24, -48, -96, \dots$

(b) $0, 1, 1, 2, 3, 5, 8, 13, 21, \dots$ and $1, 0, 1, 1, 2, 3, 5, 8, \dots$

(c) $4(1/5)^k + 7(1/3)^k, k \in \mathbb{Z}$.

2. (1p) We consider the IVP $y'' - 2xy' + 4y = 0, y(0) = 3, y'(0) = 0$ (the unknown is denoted by $y(x)$). Prove that the solution of this IVP is a polynomial.

3. (2p=0.75+0.25+0.5+0.5) We consider the scalar dynamical system $\dot{x} = -x + 2x^2 - x^3$. Represent the phase portrait. List the orbits. Specify the properties of $\varphi(t, 0)$ and $\varphi(t, 3)$. If there is an attractor, find its basin of attraction.

4. (a) (1.5p=0.5+0.5+0.5) Specify the type and stability of the linear system $\dot{x} = x, \dot{y} = -4y$. Find a global first integral $H : \mathbb{R}^2 \rightarrow \mathbb{R}$ of this system. Check that H is indeed a global first integral using the definition.

5. (a) (0.5p) Represent in the complex plane the curve $\{-2ie^{-2t+4it} : t \geq 0\}$.

(b) (2p) Find the solution of the IVP $\dot{x} = -2x - 4y, \dot{y} = 4x - 2y, x(0) = 0, y(0) = -2$. Represent in the phase plane the corresponding orbit.

(c) (0.5p) Specify the type and stability of the linear system $\dot{x} = -2x - 4y, \dot{y} = 4x - 2y$. If this system has a global first integral, find it.

(d) (0.5p) Find $\lim_{t \rightarrow \infty} e^{At}$, where A is the matrix of the system from (c).